Università degli studi di Milano-Bicocca

Data Science Lab for Smart Cities

Final Essay

Spatial and Temporal Patterns of Pedestrian Safety in Berlin: A Data-Driven Analysis of Pedestrian-Involved Traffic Accidents and Policy Implications for Smart Urban Mobility

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Abstract

This study analyses pedestrian safety in Berlin using open road traffic accident data from 2020 to 2024. A total of 7,931 pedestrian-involved accidents were examined through temporal analysis, severity analysis, grid-based spatial hotspot mapping, LOR-level demographic context analysis, and a machine learning classification model. Results show that accidents are spatially concentrated in central urban areas, that darkness and heavy vehicle involvement are associated with higher serious/fatal shares, and that temporal patterns differ between overall accident frequency and hotspot severity profiles. A Random Forest classifier achieved 60.8% accuracy in predicting serious or fatal outcomes, with hour of day, month, and day of week as the most important features. The findings support targeted safety audits, lighting improvements, heavy-vehicle risk management, and equity-aware monitoring as policy priorities.

Contents

1 Problem Description and Indicators

1.1 Introduction

Pedestrian safety in smart cities depends on the combination of technological innovation with social and policy objectives. Smart mobility strategies should give priority to safety, accessibility and environmental outcomes, rather than focus only on traffic management or digital solutions [1][2]. Intelligent transport systems and smart city technologies can help monitor risks, detect dangerous conditions and support urban mobility management [2]. However, data-driven analysis must be interpreted carefully, because official traffic accident datasets only capture reported crashes and exclude near-misses, views of safety, pedestrian exposure and everyday walking experiences [3][4][5].

At the global level, road traffic crashes remain a major public health and policy concern, with the World Health Organization reporting approximately 1.19 million road traffic deaths a year [6]. In this context, pedestrian safety is not only a technical road safety issue, but also an element of broader debates on sustainable mobility, city accessibility and social inclusion [1][6].

Pedestrian safety in Berlin will be studied through a data-driven approach. The research is based on Berlin road traffic accident data from 2020 to 2024, merged into a single dataset and filtered to pedestrian-involved accidents. The dataset includes temporal variables, accident severity, accident type, lighting conditions, involved transport modes, pedestrian involvement, and geographic coordinates. Road-surface condition was not retained as a central variable because it was not consistently available across all yearly files. The main purpose of the research is to identify spatial and temporal patterns related to pedestrian accidents and to develop relevant policy recommendations for safer and more intelligent urban mobility.

1.2 Theoretical and Literature Background

1.2.1 Smart Cities, Sustainable Mobility, and Pedestrian Safety

Smart-city strategies in the mobility sector regularly employ intelligent transport systems, data analysis, sensors, and digital platforms to improve transport management and road-user safety [2]. These approaches include real-time monitoring, evidence-based decision-making, and digital tools for managing mobility and road safety [2]. However, a people-centered smart city should regard technology as a means to achieve greater safety, accessibility, sustainability, and inclusivity, rather than as an end in itself [1][2].

Sustainable mobility frameworks emphasize that transportation systems should create access to opportunities rather than simply facilitate vehicle movement [1]. SDG 11.2.1 is especially important in this case because it shows that convenient access to public transport often requires pedestrian access; most individuals walk to and from transit stops or stations [1]. In other words, pedestrian safety is an essential aspect of sustainable mobility.

Transportation also serves a broader social function. Under the SDG framework, transportation is understood as an enabler of economic activity and social inclusion because it shapes access to employment, education, health care, markets, and public services [1]. This is especially important for individuals whose mobility options are restricted because of age, disability, poverty, or dependence on walking and public transport [1]. Pedestrian safety, therefore, forms an important component of the social fabric of a city.

In this project, the smart-city dimension is operationalized through open accident data, spatial visualization, and measurable safety indicators rather than through real-time sensor systems. Accident data can provide insights into the spatial and temporal distribution of pedestrian crashes, while spatial analysis, visualization and modelling might support decision-makers in identifying risk patterns [3][4][7]. Although studies have found that accident data alone are inadequate for measuring pedestrian safety, Schneider et al. show that locations perceived as dangerous by pedestrians may not coincide with police accident records, while Thomas et al. stress the need for additional data for a comprehensive study, such as pedestrian and vehicle volumes, land use, and socio-economic factors [3][4].

1.2.2 Pedestrian Vulnerability and the Physical Environment

Pedestrian safety is strongly influenced by the physical environment. According to Stoker et al., pedestrian injuries and fatalities are connected to high traffic volume, excessive speed, poor lighting, and urban development patterns [8]. Furthermore, Stoker et al. identify visibility, pedestrian interaction with traffic, and traffic speed as important factors contributing to pedestrian vulnerability [8]. Pedestrian accidents should therefore not only be analyzed in terms of individual responsibility, but should also be understood in relation to the physical environment [8][9].

Ewing and Dumbaugh provide a useful framework for understanding how the built environment affects traffic safety. They argue that development patterns and roadway design influence safety mainly through traffic volume and traffic speed [9]. In their framework, traffic volume is closely associated with crash frequency, while traffic speed is closely associated with crash severity [9]. This distinction is important for the present study because areas with greater pedestrian and vehicle activity may show higher accident counts, while environments with higher speeds or more severe vehicle interactions may result in more serious consequences when crashes occur [9].

The European Road Safety Observatory also identifies pedestrian safety as a specific road-safety issue requiring targeted interventions [10]. Its thematic report discusses countermeasures such as land-use planning, speed management, safe walking routes, footpaths, crossing facilities, visibility, lighting, education, enforcement, and vehicle design [10]. These measures show that pedestrian safety depends on a combination of infrastructure, regulation, vehicle technology, and user behavior [10].

A study by Kopsacheilis and Politis confirms the relevance of these factors in the local context of Berlin [11]. Their analysis of 3,257 pedestrian-involved accidents in Berlin during 2018 and 2019 used binary logistic regression and artificial neural networks to examine determinants of accident severity [11]. The results show that accident severity is associated with vehicle type, lighting conditions, speed limit, accident type, and traffic characteristics [11]. Specifically, heavy vehicle involvement and low-light conditions are linked to greater severity, while bicycle involvement and road sections with speed limits of 30 km/h or lower are associated with less serious outcomes compared to the reference categories used in the model [11].

These findings guide the selection of indicators for this study. The dataset contains variables consistent with those identified in the literature, including accident severity, accident type, lighting conditions, involved transport modes, pedestrian involvement, temporal information, and spatial coordinates [7]. Road-surface condition was not retained in the final multi-year analysis because it was not consistently available across all yearly files [7]. However, the dataset lacks several built-environment and exposure variables, such as pedestrian volume, traffic volume, sidewalk quality, detailed crossing design, speed limit, income, age, disability, and perceived safety [4][7]. This limitation should be considered when interpreting the results.

1.2.3 Spatial and Data-Driven Approaches to Pedestrian Safety

It is important to recognize that pedestrian risk varies across different parts of the city [3][4]. Schneider et al. argue that pedestrian safety planning should adopt a proactive approach rather than responding only after severe crashes have occurred [3]. Their study combines police-reported crash data with perceived-risk survey data and demonstrates that locations perceived as dangerous by pedestrians do not necessarily overlap with recorded crash locations [3]. This means that while official accident data can be useful for identifying documented accident patterns, some potentially dangerous places may remain unnoticed.

Schneider et al. show that spatial analysis and regression models can be used to evaluate the impact of exposure, roadway geometry, sidewalks, crosswalk density, and land use on pedestrian crash risk [3]. These examples confirm the value of combining spatial and statistical approaches in pedestrian safety research [3][4]. Within the field of smart urban mobility, these approaches make it possible to move from general safety concerns toward geographically targeted interventions [3][4].

The systemic pedestrian safety approach developed by Thomas et al. is also relevant to this study [4]. Their report defines a systemic approach as "data-driven, network-wide," indicating that safety analysis should address not only sites with previous crashes but also locations with similar roadway or environmental risk characteristics [4]. This approach is useful because relying only on past crash counts can overlook locations where risk factors are present but accidents have not yet been recorded [4].

Thomas et al. also emphasize that robust pedestrian safety analysis ideally requires multiple types of data, including roadway inventory, motorized traffic, pedestrian traffic, transit, land use, and socio-economic data [4]. When direct pedestrian or traffic volume data are missing, surrogate measures such as land use, transit, and demographic data may help approximate exposure, although their accuracy is uncertain [4]. This point is especially important for the present project because the Berlin dataset provides detailed accident records, but it does not directly measure pedestrian exposure or traffic volume [4][7].

1.2.4 Perceived Safety and Contemporary Mobility Conflicts

Pedestrian safety is not limited to recorded accidents; it also includes perception and experience within walking environments [3][5]. Research conducted by Kweon et al. indicates that physical features of pedestrian environments, such as sidewalk networks, landscape buffers, and street trees, may affect both walking behavior and the perception of pedestrian safety [5]. This is important because an area can be unappealing for walking or may feel unsafe despite not being identified as a hotspot for accidents [3][5].

Contemporary urban mobility also creates new conflicts between pedestrians and other modes of transportation. In addition to cars and heavy vehicles, pedestrians increasingly interact with bicycles, micromobility devices, and shared mobility systems in dense urban environments [11][12]. Shin and Choo's study on micromobility-pedestrian accidents shows that built-environment factors such as intersections, sidewalks, and subway entrances can influence where micromobility-pedestrian conflicts occur [12].

The dataset enables this project to distinguish several transport modes, including cars, bicycles, motorcycles, heavy goods vehicles, and others [7]. However, it does not detail all contemporary mobility categories. For example, IstSonstige groups together other transport modes, such as buses and trams, which means that pedestrian accidents involving trams cannot be analyzed separately without additional data [7]. This is why the present analysis focuses on the transport-mode categories that are available in the dataset and treats more detailed modal distinctions as a limitation.

1.3 Indicators for Measuring Pedestrian Safety

Measuring pedestrian safety requires indicators that capture both the frequency of accidents and the severity of their consequences. Raw accident counts are useful because they show where and when pedestrian-involved crashes are recorded. However, accident counts alone can be misleading because areas with more pedestrian activity may naturally have more accidents even if individual risk is not higher [4]. For this reason, Thomas et al. recommend considering exposure-related data, such as pedestrian and traffic volumes, when available, or using surrogate measures, such as land use, transit, or demographic information, when direct exposure data are unavailable [4].

The first indicator used in this project is pedestrian-involved accident frequency. This is measured by counting accidents in which the variable IstFuss indicates the involvement of at least one pedestrian [7]. This indicator helps identify the overall spatial and temporal distribution of pedestrian-related crashes in Berlin. Its main limitation is that it measures recorded accidents, not all risky pedestrian experiences, near-misses, or perceived danger [3][7].

The second indicator is accident severity. The variable UKATEGORIE classifies accidents according to the most severe consequence: fatal injury, serious injury, or slight injury [7]. This indicator is important because not all accidents have the same social and policy significance. A location with fewer but more severe accidents may require urgent intervention even if its total accident count is lower than that of other areas [11][4].

The third group of indicators captures temporal patterns. Variables such as UMONAT, USTUNDE, and UWOCHENTAG allow accidents to be analyzed by month, hour, and weekday [7]. These variables help identify whether pedestrian-involved accidents are concentrated during peak hours, evenings, weekends, or specific months. The main limitation is that the dataset does not include pedestrian or vehicular traffic volumes at different times; thus, it is impossible to interpret such temporal trends in terms of risks per exposure [4][7].

The fourth set of indicators concerns accident conditions and types of transportation. Variables such as ULICHTVERH, IstPKW, IstRad, IstKrad, IstGkfz, and IstSonstige are used to examine lighting conditions and the involvement of different transport modes. This is important because the literature review shows that lighting, type of vehicle, speed, and accident type are significant determinants of accident severity [11][8][10]. Nonetheless, IstSonstige includes other transportation methods as well, for example, buses and trams, and therefore, the identification of pedestrian accidents involving trams is not possible here [7].

The fifth indicator is spatial concentration. The accident dataset contains geographical coordinates, which enable mapping, grid-based hotspot analysis, and spatial joining with LOR Planungsräume. Spatial indicators help to pinpoint accident hotspots and suggest geographically specific policies [3][4]. The main limitation is that spatial concentration does not necessarily mean higher personal exposure without taking into account pedestrian activity, traffic density, or population statistics [4].

These indicators were selected because they can be measured using the available dataset, are connected to the literature, and are relevant for decision-making. These indicators allow the project to describe where and when pedestrian collisions happen, evaluate their severity, and determine what kinds of accident situations are more likely to involve pedestrians [11][4][7].

1.4 Ethical and Social Implications

The study of pedestrian safety from a data-driven perspective raises ethical and social considerations because accident data provide only a partial representation of pedestrian safety. Although official accident data contain information about actual accidents, they do not contain information about near-misses, perceptions of risk, or areas where pedestrians change their behavior due to feelings of danger [3]. Schneider et al. note the discrepancy between perceived and police-reported risky locations, and thus, any policy solely based on official crash data may prove insufficient in tackling pedestrian risk in some areas [3].

The available Berlin dataset also has important social limitations because it does not include detailed information about the pedestrians involved, such as age, gender, disability status, income, or migration background

[7]. Therefore, the current data do not allow this project to identify which social groups are more exposed to pedestrian risk. Nonetheless, the SDGs, alongside research on pedestrian safety, emphasize the need to give special consideration to vulnerable groups, such as elderly people, children, and disabled individuals [8][1].

The issue of equity must be considered when choosing the right indicators as well. When relying only on the number of accidents, places with more incidents may receive priority, while others with fewer reports, fewer pedestrians, or higher perceived danger may be neglected [3][4]. Moreover, without considering pedestrian exposure, it becomes challenging to differentiate between dangerous locations because of poor infrastructure and places with high traffic [4]. Therefore, the findings should be interpreted as evidence of reported pedestrian-involved accidents, not as a comprehensive measure of pedestrian safety or urban justice.

Finally, smart-city technologies for pedestrian safety require careful governance. Tools such as intelligent transport systems, sensors, and cameras could help identify and address dangers, but at the same time, they raise issues related to privacy, surveillance, unequal implementation, and responsibility [2]. Sosik and Iwan discuss several smart-city solutions designed to improve pedestrian safety [2]. However, the technologies used for pedestrian safety should not just demonstrate good results but also be socially acceptable and transparent [2]. A safer smart city will thus require both data analysis and inclusive planning [1][2].

2 Data Analytics, Optimization and Policy Suggestions

2.1 Data and Methodology

2.1.1 Dataset Description

This study uses the empirical analysis of road traffic accident data in Berlin during the period 2020–2024. The original datasets containing yearly road accidents in Germany have been compiled into a harmonized dataset. Only those accidents involving pedestrians, defined by the attribute IstFuss, were included in this study. The total number of pedestrian accidents in Berlin from 2020 to 2024 amounted to 7,931.

The data set contains temporally oriented variables, including Year, Month, Hour, and Weekday, accident outcome variables, like Accident Severity, accident context variables like Lighting and Accident Type, transport mode variables, denoting whether a car, bicycle, motorbike, HGVs, or any other transportation mode is involved, and spatially oriented variables denoting the coordinates of the accidents. The Severity variable UKATEGORIE classifies accidents into three groups: fatal, serious, and slight injuries.

2.1.2 Data Cleaning

First, the unprocessed files were read year by year. The column names of the file for 2021 were modified since this file differed somewhat in structure compared to the others. Thus, only accidents that occurred in Berlin were selected based on the state-federal code. Coordinates were converted to numeric values by replacing commas with periods.

It was decided to retain only the variables present in all years for further analysis. As such, the road-surface condition variable was excluded from the central variables, as this information varied across annual files. After merging the five years, the dataset was filtered to retain only pedestrian accidents. Labels were additionally added for severity, lighting conditions, days of the week, month, and accident type.

2.1.3 Analytical Strategy

The study involved two steps. Descriptive and severity analyses were used to investigate the characteristics of pedestrian accidents by year, month, time of day, weekday, lighting conditions, and transport mode. The key severity measure was the proportion of fatal or serious-injury outcomes among all reported pedestrian involved accident cases.

The second approach adopted was a grid-based hotspot analysis. The accident locations were first projected to the projected coordinate system and then placed in a 500m $\times$ 500m grid. The number of pedestrian accidents, the number of serious/fatal accidents, the severity rate, and the percentages of accidents that occurred during dark periods, with motor cars, bicycles, motorcycles, and HGVs were computed for each grid cell.

The approach focuses on detecting spatial hotspots of recorded pedestrian accidents. Nevertheless, the technique does not provide a true measure of the probability of an accident, since the analysis is restricted to recorded accidents and lacks information on pedestrian presence, traffic density, and non-accident areas. The spatial hotspots generated by the analysis should be understood as clusters of reported accidents only [3][4].

2.1.4 Contextual LOR Demographic Analysis

To extend the hotspot analysis, the project also includes a contextual demographic analysis at the level of Berlin LOR Planungsräume. The purpose of this step is to examine whether areas with high concentrations of pedestrian-involved accidents also have distinctive residential demographic characteristics, such as higher shares of children, elderly residents, women, or higher population density.

The analysis uses the LOR Planungsräume boundaries valid from 2021 as a common spatial reference for the accident data from 2020 to 2024 [13]. The accident counts were then aggregated by LOR area and merged with 2024 demographic data for Berlin LOR Planungsräume [14].

Several contextual indicators were created for each LOR area. These include total pedestrian accident count, serious/fatal accident count, accidents per 10,000 residents, serious/fatal accidents per 10,000 residents, population density, share of children aged 0–17, share of elderly residents aged 65 and over, and female share. These indicators make it possible to compare accident concentration with the demographic profile of the surrounding area.

This analysis is contextual rather than causal. It does not identify the age, gender, health status, or disability status of the pedestrians involved in accidents. Instead, it describes the demographic composition of the areas where reported pedestrian-involved accidents are more concentrated. Therefore, the results should be interpreted as area-level associations, not as individual-level evidence about which demographic groups are more likely to be involved in accidents.

2.1.5 Hotspot Priority Classification

To derive priority classifications for hotspots from the spatial analysis and demographic profiling undertaken earlier, it is important to understand how to use these data in a practical sense to make meaningful classifications for hotspot prioritization. As the dataset is comprised only of locations where accidents occurred rather than all other places, it would not be useful to classify LORs based on where accidents may occur in the future. Instead, the model was used to categorize LORs based on two criteria: accident rate and seriousness. LORs categorized as high in either the number of accidents or proportion of serious/fatal accidents are considered high seriousness, and those categorized as high count received high count classification.

2.1.6 Classification Model

Beyond the descriptive and spatial analyses, a classification model was built to explore whether it is possible to predict the severity of a pedestrian accident from the information available in the dataset. The idea was straightforward – given what we know about when an accident happened, what lighting conditions were present, and which vehicles were involved, can we predict whether the outcome would be serious or fatal rather than minor?

The target variable was binary: 1 for fatal or serious injury, 0 for slight injury. Before training the models, four additional features were created from the existing variables. A night indicator was added to flag accidents that happened between 22:00 and 05:00. A weekend indicator was created from the day of week variable. A dangerous vehicle indicator was built by combining heavy goods vehicle and motorbike involvement into a single flag. Finally, a dark and night indicator combined the darkness lighting condition with the night time indicator to capture the overlap between the two risk factors.

Two models were trained and compared – Logistic Regression and Random Forest. Both were trained on 80% of the data and tested on the remaining 20%, using stratified sampling so that the proportion of serious and fatal cases was preserved in both sets. Because the dataset is imbalanced – roughly 76% of accidents resulted in slight injuries and only 24% in serious or fatal outcomes – both models used balanced class weights to avoid simply predicting the majority class every time.

2.2 Results

2.2.1 Descriptive Overview

This is evident in the dataset, which records 7,931 pedestrian accidents in Berlin from 2020 to 2024. For the period between 2020 and 2024, there were 1,396 recorded pedestrian-involved accidents in 2020, 1,460 in 2021, 1,569 in 2022, 1,794 in 2023, and 1,712 in 2024.

Figure 1 shows that reported pedestrian-involved accidents increased after 2020, reached the highest value in 2023, and slightly decreased in 2024.

Regarding the severity of the accidents, the dataset consists of 6,034 slight injury accidents, 1,825 serious injury accidents, and 72 fatal accidents. In this way, it is evident that most pedestrian accidents have resulted in minor injuries; however, serious accidents have occurred, and some have been fatal.

2.2.2 Severity by Lighting Conditions

Lighting conditions are associated with differences in the share of serious or fatal outcomes. The highest percentage of serious-to-fatal outcomes occurred in accidents that occurred in the dark, at 27.8%. At dusk and dawn, the percentage was 23.8%, compared to only 22.3% at daylight.

The above results are consistent with the literature, which highlights visibility as a key aspect of pedestrian safety [8][10]. It is also corroborated by the findings from the Berlin region, which have indicated a relationship between lighting conditions and the severity of pedestrian accidents [11]. Nevertheless, the above relationship cannot be taken to imply causality owing to the lack of other contextual variables such as vehicle speeds and exposure.

2.2.3 Severity by Transport Mode

The most notable difference in severity lies among the transport modes involved. Those involving heavy goods vehicles showed the largest portion of incidents with fatalities or serious injuries, namely 45.9%. Accidents

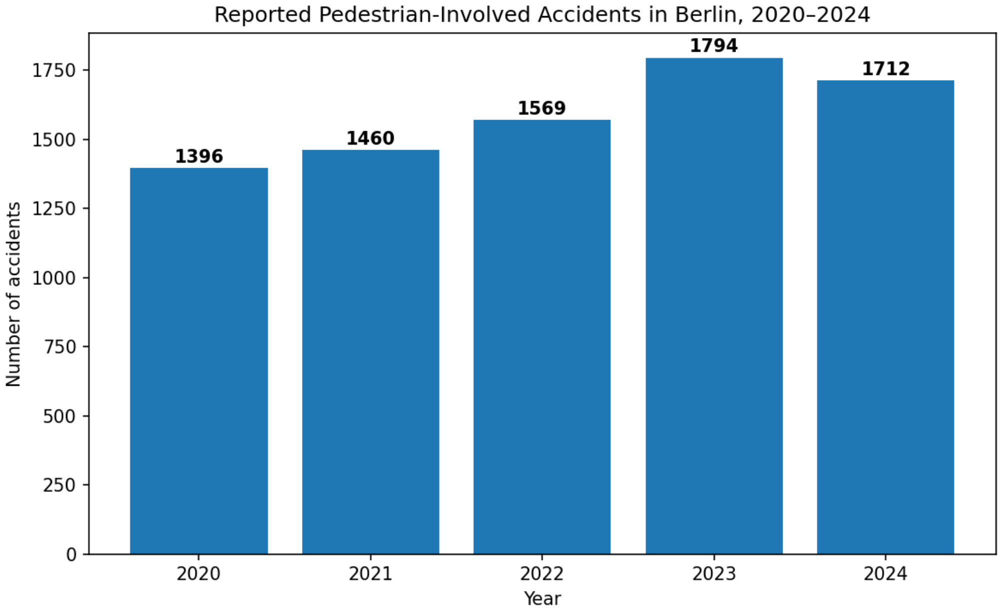

Figure 1: Pedestrian-involved accidents in Berlin by year, 2020–2024.

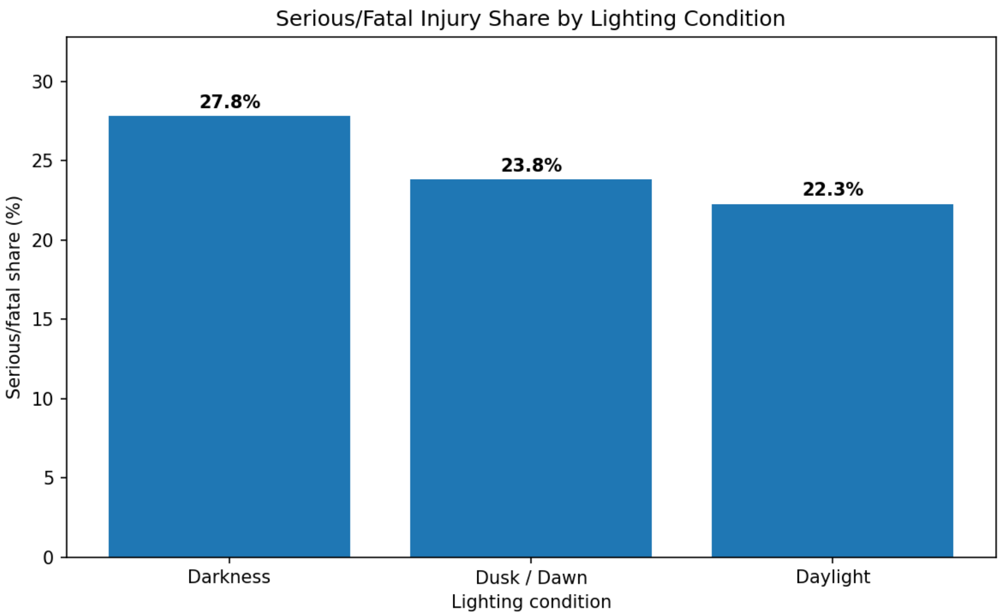

Figure 2: Percentage of serious or fatal injuries by lighting condition, Berlin 2020–2024.

involving cars had a serious/fatal share of 26.2%. Likewise, accidents involving motorcycles had a serious/fatal share of 26.1%, whereas incidents involving bicycles accounted for 15.6%.

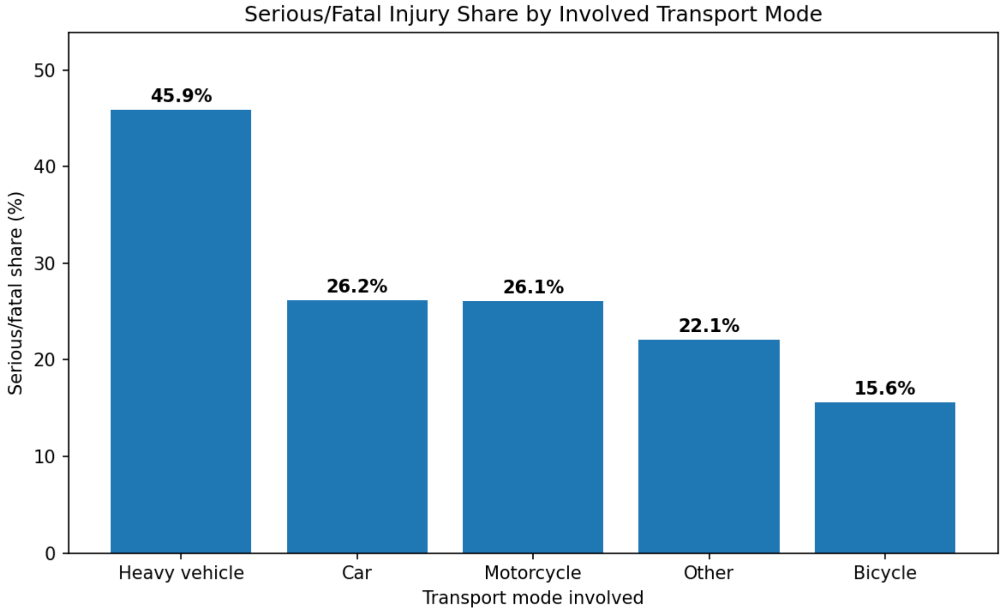

Figure 3: Percentage of serious or fatal injuries by vehicle type, Berlin 2020–2024.

This finding supports the importance of vehicle type in pedestrian accident outcomes. It is in agreement with the research carried out by Kopsacheilis and Politis in Berlin, where heavy vehicles were associated with a greater level of severity of pedestrian accidents, whereas bicycles contributed towards a lower level of severity than the benchmark values in their research model [11]. Regarding policies for preventing severe pedestrian accidents, these findings suggest an urgent need to focus on areas where pedestrians interact with heavy vehicles.

2.2.4 Temporal Patterns

The hourly analysis of the relationship between pedestrian accident severity shows that the highest percentages of serious and fatal accidents are recorded during late night and early morning hours. Specifically, the highest percentages are observed for 02:00, 06:00, 01:00, 23:00, and 00:00.

Nevertheless, it does not imply that these hours are the most dangerous since there is no data on pedestrian traffic by hour. However, a different trend can be observed in the time series of the hot spot. In the grid cells that belong to the top accident hot spots, more incidents occurred during the afternoon and early evening period at about 16:00, 17:00, and 18:00. This indicates that perhaps the occurrence of accidents within hot spots could depend on high urban activities. As usual, this statement must be taken carefully since data on pedestrians' and vehicles' traffic is not available.

2.2.5 Spatial Hotspot Analysis

The spatial hotspot analysis was performed using grid cells of size 500 metres by 500 metres. In total, within the city of Berlin, there were 1,602 grid cells that each had at least one recorded pedestrian accident.

The grid that recorded the largest number of pedestrian crashes from 2020 to 2024 is grid 767, which has 50 accidents in total. Grid 767 also records 16 major/serious accidents, out of which there was one death, and the major share is 32%. Among other grids with large numbers of incidents are grid 2751, which records 40 accidents, and grids 2203 and 2333, recording 35 accidents each.

However, from the analysis of hotspots where accidents become fatal, it is evident that those places where the accidents are serious do not necessarily have the largest numbers or proportions of fatalities. In this regard, grid 2889 recorded only 16 accidents overall but 9 serious or fatal accidents, thus generating an overall proportion of serious/fatal accidents as 56.25%.

Therefore, the focus should not only be on the total number of accidents in any area, but rather the serious/fatal share in the areas too.

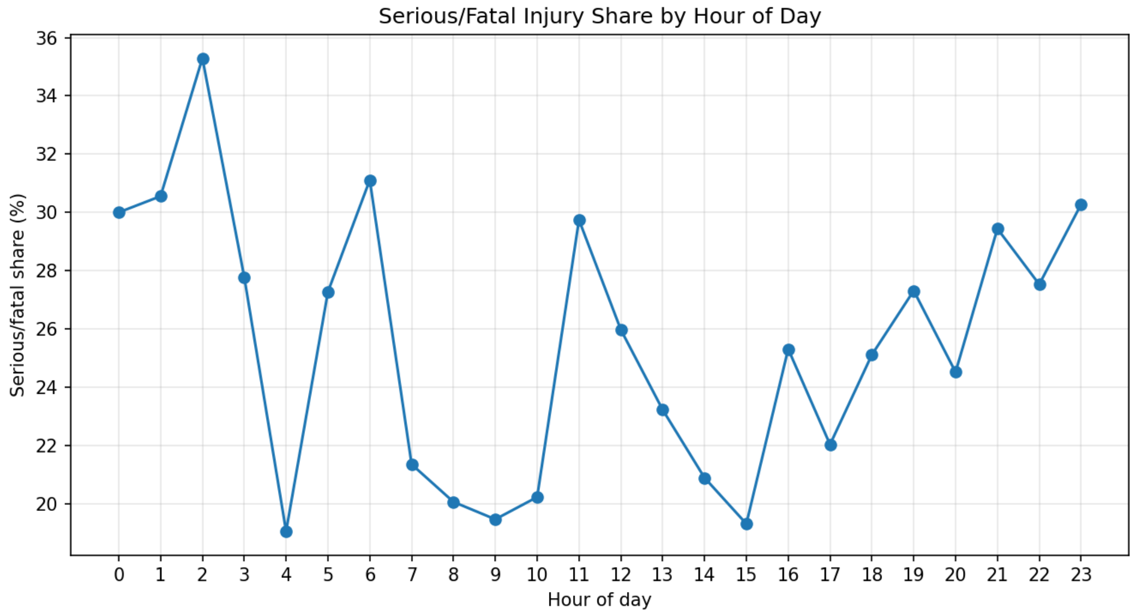

Figure 4: Percentage of serious or fatal injuries by hour of day, Berlin 2020–2024.

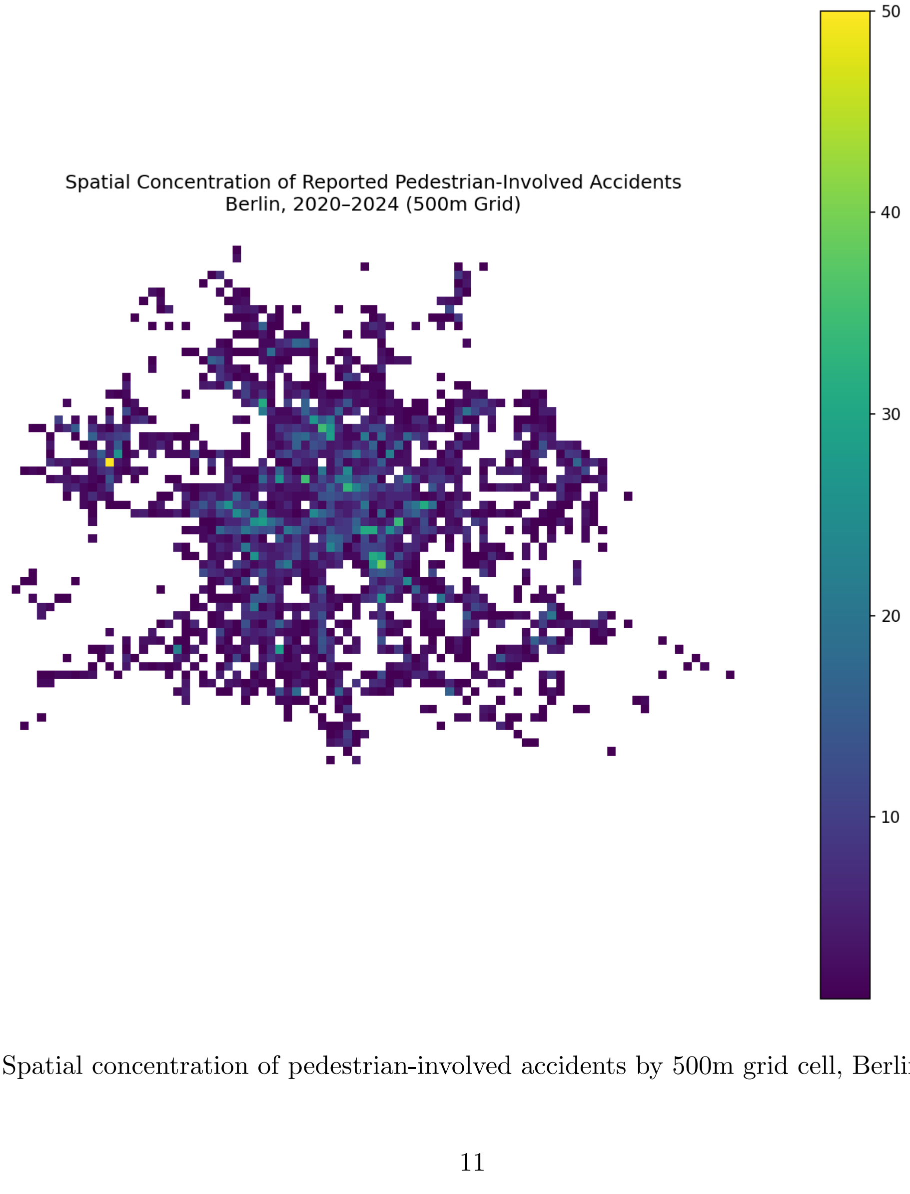

Figure 5: Spatial concentration of pedestrian-involved accidents by 500m grid cell, Berlin 2020–2024.

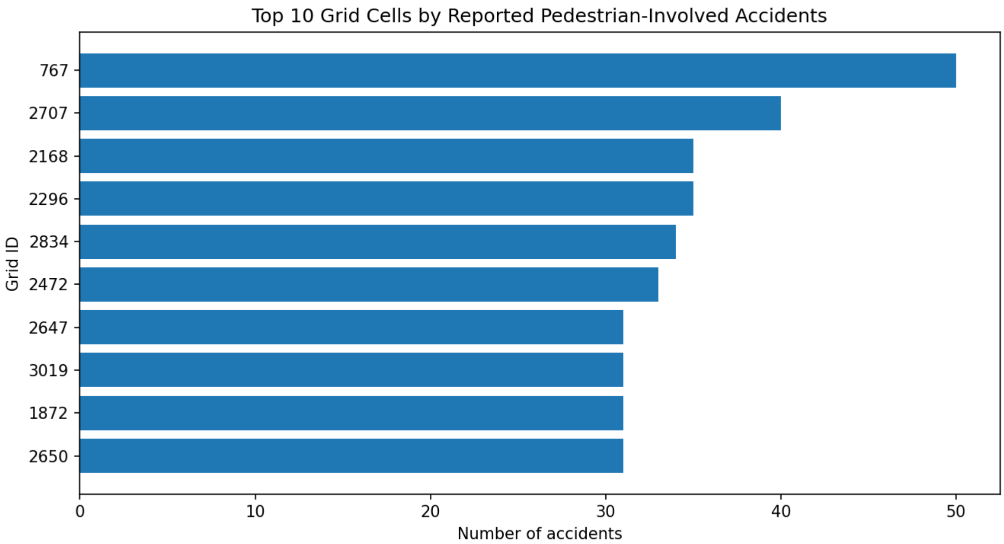

Figure 6: Top 10 Grid Cells by Reported Pedestrian-Involved Accidents

2.2.6 Hotspot Profiles

Comparison of the accident hotspot grids in the top ten and all other grids sheds additional light on the matter. The accident hotspot grids in the top ten featured 351 accidents in total, accounting for roughly 4.4% of all pedestrian accidents in the database.

	Area type	Total accidents	Serious/fatal share	Darkness share	Heavy vehicle share	Car share	Bicycle share
0	Top 10 accident hotspot grids	351	21.08%	27.92%	0.28%	53.56%	26.21%
1	All other grids	7580	24.05%	27.90%	1.42%	65.44%	17.04%

Figure 7: Comparison of top-ten accident hotspot grids and all other grids by accident count, severity share, darkness share, and vehicle involvement.

The ratio of serious or fatal accidents among those 351 was at 21.08%, which was less than the rate of 24.05% seen in all other grids.

The distribution of accidents also revealed that the percentage of accidents that took place under dark conditions was about equal in both hotspot and non-hotspot zones, with about 28% in each category. Heavy vehicle involvement was more common outside the top accident hotspots compared to within them. In contrast, bicycle involvement was more common in the hotspot areas. This could be interpreted as meaning that the key hotspots for the number of accidents were primarily urban locations where both pedestrians and bicycles are common, rather than places where heavy vehicles dominate. However, this interpretation remains limited because the dataset does not include pedestrian volume, bicycle volume, road hierarchy, street design, or public transport flow.

2.2.7 Demographic Context of Accident Hotspots

From the LOR analysis, the regions where the number of pedestrian-related accidents is high comprise mostly central, highly active urban centers. The highest recorded number of pedestrian-related accidents was from Alexanderplatzviertel with a total number of 93 accidents between 2020 and 2024. Some other highly populated LOR areas include Gesundbrunnen with 84 accidents, Oranienburger Straße with 83 accidents, Carl-Schurz-Straße with 75 accidents, Breitscheidplatz with 74 accidents, Charitéviertel with 67 accidents, and Unter den Linden with 66 accidents.

These regions are not simply residential communities, which have many vulnerable individuals. Some of these regions are central city zones, which perform vital functions of commerce, tourism, employment, public transportation, or institutions. It implies that the accident data will not only represent the resident population of such a region, but also the pedestrian activity, which is associated with non-resident individuals.

Top LOR areas by total pedestrian accidents:

	PLR_ID	total_accidents	severe_accidents	fatal_accidents	serious_accidents	minor_accidents	dark_accidents	car_accidents	bicycle_accidents	motorcycle_accidents	heavy_vehicle_accidents	other_mode_accidents	severe_rate	dark_share	car_share	bicycle_share	heavy_vehicle_share
9	01100310	93	23	0	23	70	28	41	23	3	1	26	0.247312	0.301075	0.440860	0.247312	0.010753
32	01300733	84	17	0	17	67	25	50	21	6	1	8	0.202381	0.297619	0.595238	0.250000	0.011905
8	01100309	83	14	0	14	69	23	41	27	5	1	12	0.168675	0.277108	0.493976	0.325301	0.012048
210	05100316	75	17	2	15	58	21	53	8	0	0	16	0.226667	0.280000	0.706667	0.106667	0.000000
184	04501042	74	8	0	8	66	28	45	12	4	0	14	0.108108	0.378378	0.608108	0.162162	0.000000
7	01100308	67	18	0	18	49	19	34	19	3	0	13	0.268657	0.283582	0.507463	0.283582	0.000000
5	01100206	66	15	0	15	51	18	24	25	2	0	15	0.227273	0.272727	0.363636	0.378788	0.000000
35	01300836	63	10	2	8	53	27	43	8	2	2	9	0.158730	0.428571	0.682540	0.126984	0.031746
500	12200309	62	16	1	15	46	10	45	10	2	1	5	0.258065	0.161290	0.725806	0.161290	0.016129
221	05200629	58	16	0	16	42	13	37	6	3	0	12	0.275862	0.224138	0.637931	0.103448	0.000000

Figure 8: Top LOR areas by reported pedestrian-involved accident count, Berlin 2020–2024.

When accident counts are normalized by residential population, several central areas show very high accident rates per 10,000 residents. For example, areas such as Unter den Linden, Großer Tiergarten, Breitscheidplatz, Charitéviertel, and Alexanderplatzviertel have high accident rates relative to their resident population. However, this indicator should be interpreted carefully, because central areas may have relatively small residential populations but very high daytime pedestrian flows. Therefore, accidents per 10,000 residents should not be interpreted as direct pedestrian risk. It is better understood as a contextual indicator showing where reported accident concentration is high relative to the number of people living in the area.

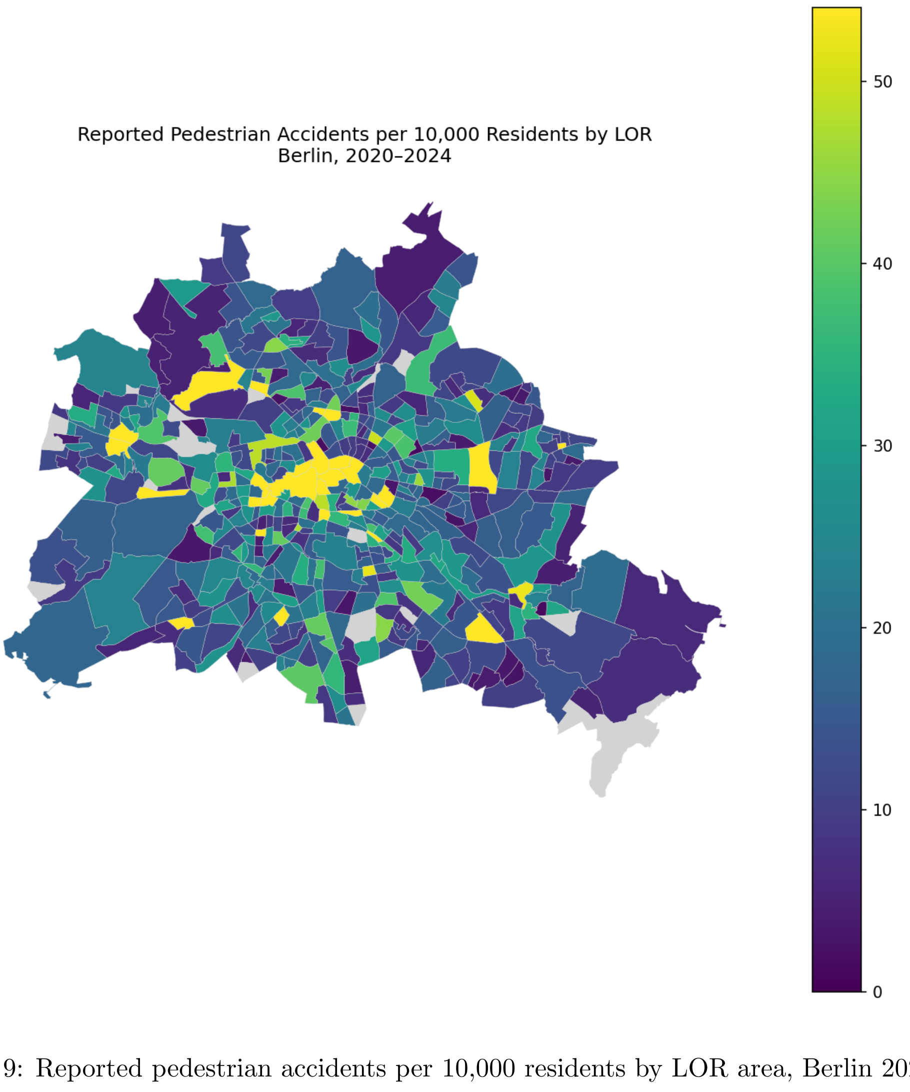

Figure 9: Reported pedestrian accidents per 10,000 residents by LOR area, Berlin 2020–2024.

The comparative analysis of the top ten LORs in terms of accidents and the remaining LORs allows us to understand their social implications.

Firstly, the accident rate in the top ten LORs according to the number of accidents was significantly higher compared to the number of residents living in the other LORs in Berlin. However, there was no significant difference in the proportion of children and elderly people living in the LOR areas with a larger number of accidents.

Top 10 LOR areas by total pedestrian accidents:

	PLR_ID	PLR_NAME	borough_name	total_accidents	severe_accidents	severe_rate	population_total	population_density	accidents_per_10k_residents	children_share	elderly_share	female_share
424	01100310	Alexanderplatzviertel	Mitte	93	23	24.73	10181.0	7457.0	91.35	12.03	14.59	47.19
336	01300733	Gesundbrunnen	Mitte	84	17	20.24	11684.0	17749.0	71.89	17.17	11.17	48.38
334	01100309	Oranienburger Straße	Mitte	83	14	16.87	12651.0	12509.0	65.61	14.21	9.60	49.01
487	05100316	Carl-Schurz-Straße	Spandau	75	17	22.67	10505.0	7626.0	71.39	16.33	16.88	50.79
530	04501042	Breitscheidplatz	Charlottenburg-Wilmersdorf	74	8	10.81	3664.0	6069.0	201.97	14.44	20.66	49.59
450	01100308	Charitéviertel	Mitte	67	18	26.87	7297.0	4337.0	91.82	13.14	10.15	48.91
446	01100206	Unter den Linden	Mitte	66	15	22.73	1876.0	1446.0	351.81	12.69	13.27	48.13
308	01300836	Humboldthain Nordwest	Mitte	63	10	15.87	14947.0	10370.0	42.15	16.92	10.02	47.32
413	12200309	Scharnweberstraße	Reinickendorf	62	16	25.81	11286.0	15947.0	54.94	16.57	15.75	48.39
140	05200629	Borkumer Straße	Spandau	58	16	27.59	8357.0	6304.0	69.40	15.97	22.53	50.81

Figure 10: Demographic context comparison for the top accident-count LORs and other LOR areas.

This result reinforces the case for the need for careful interpretation of accident data. The communities with the most pedestrian accidents recorded do not have to be the same communities where there is the highest concentration of socially vulnerable people. A proper pedestrian safety analysis from an equity perspective will need more data including pedestrian flows, vehicle flows, transit flows, location of schools and workplaces, tourist presence, and pedestrian characteristics in accidents.

2.2.8 Hotspot Priority Classification

The priority ranking system makes a tangible method for translating the findings into actionable interventions. Areas that qualify as Priority A include high numbers of accidents along with high severity burden; hence, they become the best candidate areas for conducting thorough street safety evaluations. Priority B areas are characterized by high number of accidents and low severity burden; thus, they may indicate high pedestrian movements, as well as multimodal activities. Priority C areas are identified as having low numbers of accidents, yet having high severity burden; therefore, these areas tend to be involved in fewer accidents, but when they do occur, they turn out to be highly damaging.

```
High accident count threshold: 29.80000000000001
High severe accident count threshold: 8.0
High severity rate threshold: 0.333
Priority classes created:
```

priority_class	count
Priority D: lower count + lower severity	373
Priority C: lower count + high severity	116
Priority A: high count + high severity	38
Priority B: high count + lower severity	15

dtype: int64

Figure 11: Hotspot priority classification thresholds and number of LOR areas in each class.

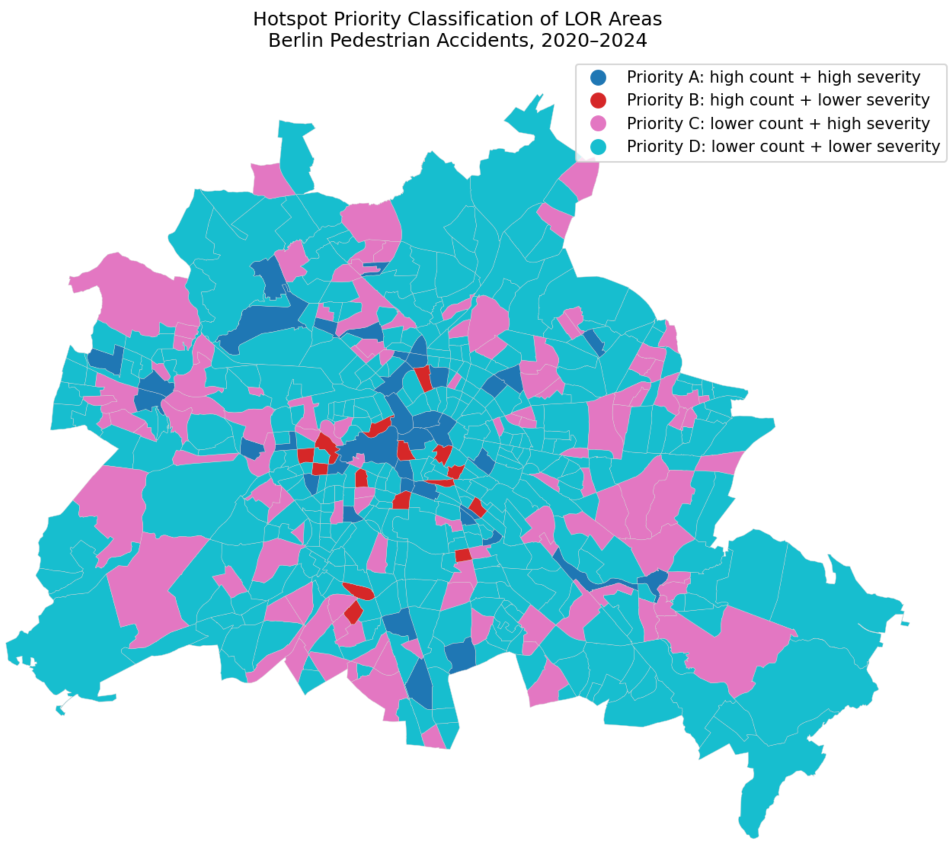

Figure 12: Hotspot priority classification of LOR areas. Priority A areas combine high accident frequency with high severity burden and should receive the strongest attention in street-level safety audits.

2.2.9 Classification Model Results

Between the two models, Random Forest performed better. It achieved a test accuracy of 60.8% and a cross-validation mean of 59.3% across five folds, with a standard deviation of 2.2%, which means its performance was consistent and not just a result of a lucky split. Logistic Regression reached a test accuracy of 52.6% and a cross-validation mean of 51.1%, which is only marginally better than random guessing given the class distribution.

Looking at the feature importance results for the Random Forest, the picture is dominated by temporal variables. Hour of day came out as the most important feature at 22.2%, followed by month at 20.8% and day of week at 14.3%. Year contributed 12.9%, which likely captures the gradual post-COVID recovery in accident volumes rather than a meaningful safety trend. Bicycle involvement came in at 6.4% and lighting at 6.2%, with the remaining vehicle type variables making up the rest.

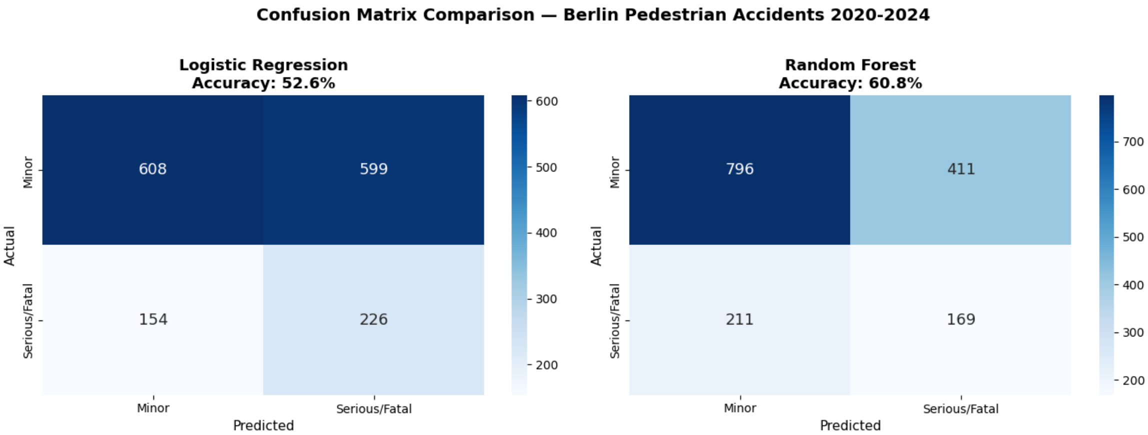

Figure 13: Confusion matrix comparison between Logistic Regression and Random Forest classifiers, Berlin 2020–2024.

Model Comparison:

	Model	Test Accuracy %	CV Mean %	CV Std %
0	Logistic Regression	52.6	51.1	3.2
1	Random Forest	60.8	59.3	2.2

Figure 14: Results of Logistic Regression and Random Forest models.

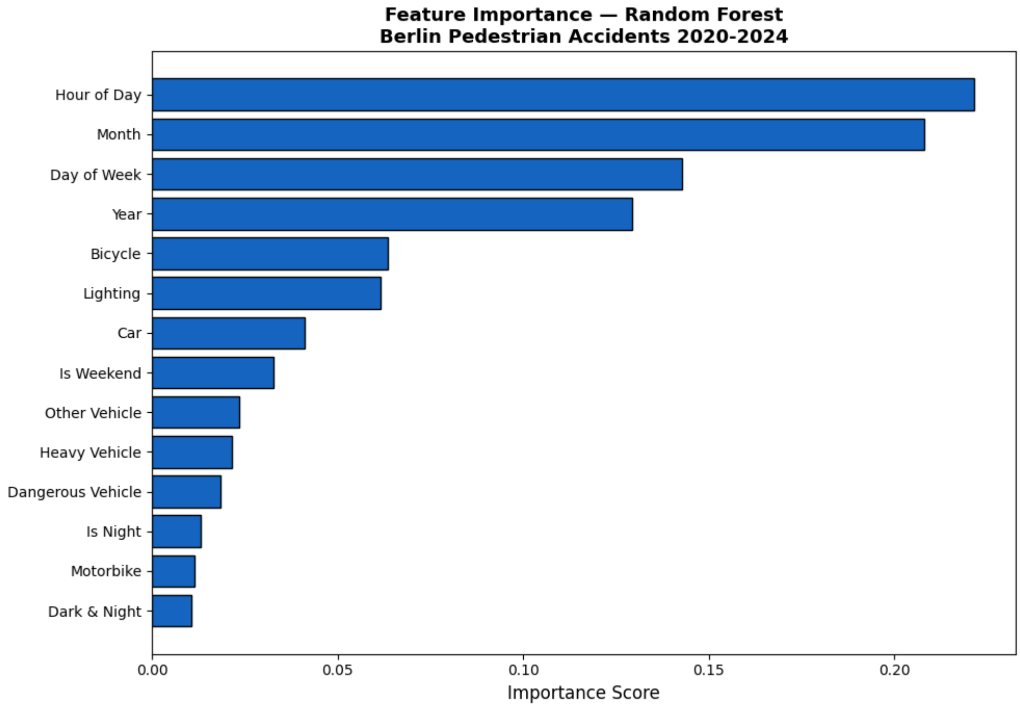

Figure 15: Feature importance scores from the Random Forest classifier.

The moderate overall accuracy is not surprising. The dataset simply does not contain the variables that the literature most consistently links to pedestrian accident severity – vehicle speed, road type, crossing design, pedestrian age, or disability status [11][4]. What the model does show is that temporal patterns carry real predictive signals. When and under what conditions an accident happens does matter for how serious it turns out to be, even if the available features cannot capture the full picture. The model is best understood as an exploratory tool rather than something that could be deployed in practice.

2.3 Discussion

These results are also supported by several theoretical insights from the existing literature. For instance, the empirical evidence obtained confirms that visibility and illumination levels play an important role in pedestrian safety. The serious/fatal proportion of accidents was higher for accidents that took place during darkness compared to those in daylight [11][8][10].

Second, the link between high proportions of heavy vehicle participation and serious/fatal outcomes in pedestrian accidents in Berlin aligns with other studies conducted in Berlin specifically. Despite accounting for just a small proportion of all pedestrian accidents, heavy vehicles were responsible for a substantially higher proportion of serious/fatal incidents. This underscores the need to consider outcome severity as well as frequency when developing pedestrian safety policies.

The third important finding is that the incidents involving pedestrians are located in specific zones of Berlin and are not equally distributed in the entire city space. Therefore, it becomes clear why an approach combining the use of space analysis and systematics is so important for increasing pedestrian safety as they help understand which zones require preventive intervention.

However, the analysis demonstrates the limitations of such an approach, namely, that accident counting alone does not help much in defining priority zones. In some cases, the zones with the lowest number of accidents turned out to be the zones with the highest proportion of severe/fatal accidents. Thus, when making policy decisions, attention must be paid to both frequency and severity of pedestrian-related accidents.

The classification model adds one more layer to this picture. The Random Forest achieved 60.8% accuracy, which is moderate but stable across cross-validation folds. What stands out from the feature importance analysis is how strongly temporal variables dominated – hour of day, month, and day of week together accounted for more than half of the model's predictive power. This is consistent with the temporal patterns identified earlier in the analysis, where late night hours and specific months showed noticeably higher serious/fatal shares. The relatively low contribution of the lighting variable on its own likely reflects the fact that darkness and late-night hours are closely correlated, meaning the model captures the lighting effect indirectly through the hour variable. The moderate overall accuracy is expected given that the dataset does not include the variables most consistently linked to severity in the literature, such as vehicle speed, road type, or pedestrian characteristics [11][4]. The model should therefore be read as an exploratory tool that confirms the relevance of temporal context rather than as a predictive system ready for deployment.

In summary, the issue of pedestrian safety in Berlin can neither be addressed nor described using just one form of indicator. Numbers of accidents, proportions of accidents involving serious injuries and fatalities, spatial locations of accidents, modes of transportation involved, lighting conditions, demographics, and the classification model results all highlight different dimensions of the same problem. The temporal variables that dominated the Random Forest model – hour of day, month, and day of week – point in the same direction as the descriptive analysis: when an accident happens shapes how serious it is likely to be, just as much as what vehicle is involved or how well-lit the road is. Taken together, these findings reinforce the case for a multi-indicator approach to pedestrian safety in Berlin, one that looks at frequency and severity together, accounts for spatial concentration, considers demographic context, and uses predictive tools not as definitive answers but as a way of asking better questions about where and when the risk is highest.

2.4 Policy Recommendations

2.4.1 Prioritize Hotspot-Based Safety Audits

The hotspot analysis based on the grid method identifies the areas that have a large number of accidents involving pedestrians. These areas are to be selected for street-level safety audits, which would include an examination of the crosswalk design, signalization, turning movement, pedestrian waiting place, visibility, illumination, and possible conflicts with bicyclists. The proposed strategy is similar to a systematic approach to pedestrian safety, which focuses on identifying the problematic locations and road conditions before more accidents happen [4].

2.4.2 Distinguish Between Accident-Count Hotspots and Severity Hotspots

It can be seen from the results that those places where there are the greatest amounts of accidents are not necessarily those where the most serious or fatal accidents take place. For this reason, two prioritization layers are recommended in Berlin. One layer should identify areas with high accident counts, while the other should identify areas with high serious/fatal counts or high severity rates.

2.4.3 Improve Lighting and Visibility

As accidents in the dark are characterized by a greater percentage of serious and fatal cases, lighting measures may be adopted if accidents take place in low-light conditions in particular. Such actions may range from better lighting of streets to enhancing visibility in crossing areas, removing visual barriers in crossing zones, and adopting reflective and visible crossing designs. All these steps conform to pedestrian safety guidelines recommended in Europe [10].

2.4.4 Manage Heavy-Vehicle Risk

It was found that the presence of heavy goods vehicles correlated with the highest share of serious/fatal cases. Therefore, locations where pedestrians are likely to encounter such types of transport need special attention regarding the adoption of relevant safety measures. Such measures could consist of limited entry of heavy goods vehicles into high-pedestrian traffic zones during certain hours, better delivery organization, pedestrian crossing redesign, and reduced speed limits. This advice coincides with the findings provided by existing research on pedestrian accidents [11].

2.4.5 Focus on Afternoon and Early-Evening Hotspot Activity

For the accident hotspots, the most common accidents occurred between 4 pm and 6 pm. This means that any attempt to control these accident hotspots would have to incorporate afternoons and early evenings. The strategies might include actions such as police patrols, changes in traffic light timing, or traffic calming during periods of heavy pedestrian flow. Nevertheless, further analysis on pedestrians' and vehicles' movements is required.

2.4.6 Use Demographic Context for Equity-Aware Prioritization

The LOR demographic analysis shows that the areas with the highest accident counts do not necessarily have higher shares of children or elderly residents. This does not mean that vulnerable groups are not important for pedestrian safety. Rather, it shows that residential demographic indicators alone cannot fully explain accident concentration in central urban areas. Berlin should therefore combine accident data with demographic indicators, pedestrian counts, public transport flows, and local street audits when prioritizing interventions. This would help avoid focusing only on central accident-count hotspots while neglecting areas where vulnerable residents may experience high perceived risk or underreported safety problems.

2.4.7 Connect Data-Driven Monitoring with Berlin's Existing Policy Framework

The recommendations are consistent with Berlin's Road Safety Programme 2030, which aims to reduce crashes involving fatalities and serious injuries and move closer to the Vision Zero approach [15]. They also align with Berlin's Mobility Act and pedestrian mobility policies, which emphasize improving conditions for people walking in the city [16]. A data-driven monitoring dashboard could help the city update hotspot locations over time, compare accident-count and severity hotspots, integrate demographic context, and evaluate whether interventions reduce serious and fatal pedestrian injuries.

2.5 Limitations

Some limitations can be noted about this research. First of all, the database contains only accidents that happened and does not contain close calls or incidents related to feeling dangerous and changes in pedestrian behavior due to unsafe conditions for pedestrians.

Second, the data set lacks information on pedestrian exposure, bike exposure, traffic flow, public transit use, tourism, or employment density. Hence, the study is unable to calculate the true probabilities of accidents happening at certain locations. The study may be able to locate zones with many accidents but will fail to evaluate pedestrian risk exposure-adjusted.

Thirdly, no individual data on the pedestrians, including age, gender, physical disability, health condition, income level, and immigrant history, is available in the dataset. Ecologically, the demographic analysis based on the LORs is derived from aggregate data, and cannot be considered an indicator of the age or gender of the injured pedestrians.

Fourth, caution is necessary in the interpretation of accident rates on a population basis. There are cases where the resident population in a particular area might be small, while the level of activity during the day is high because of commuters, tourists, shoppers, students, employees, and other users of public transport in the area.

Fifth, LOR boundaries for 2021 were utilized as a common geographical base for accident statistics from 2020 to 2024. Such a method allows ensuring the highest consistency while matching accidents with demographics indicators, though it cannot reflect the actual geographical hierarchy of 2020 precisely.

Sixth, the hotspot analysis based on grids depends on the size of the grid chosen. The use of a 500m $\times$ 500m grid offers a good compromise between accuracy and interpretation, but varying grid sizes could affect the results.

Finally, the research remains descriptive and exploratory, revealing trends but not establishing causation. In any policy implementation, this would need to be supplemented by field studies, infrastructure surveys, pedestrian and vehicular traffic counts, and community input.

2.6 Conclusion

This project analyzed pedestrian safety in Berlin using open road traffic accident data from 2020 to 2024. The analysis focused on pedestrian-involved accidents and examined temporal patterns, severity patterns, transport-mode involvement, spatial hotspot concentration, and demographic context.

The results show that pedestrian accidents are spatially concentrated in specific parts of Berlin and that severity varies by lighting condition and involved transport mode. Accidents in darkness had a higher serious/fatal share than accidents in daylight, and accidents involving heavy vehicles had the highest serious/fatal share among the analyzed transport modes. The hotspot analysis also showed that the areas with the highest accident counts are not necessarily the areas with the highest severity rates, suggesting that policy prioritization should consider both frequency and severity.

The LOR demographic analysis adds an important contextual dimension. The areas with the highest accidents count were mainly central and high-activity urban areas and did not have higher numbers of children or elderly residents than the rest of Berlin. This suggests that accident concentration in these areas may be shaped by non-residential urban activity, such as commuting, tourism, retail, institutional activity, and use of public transport. However, because the dataset does not include pedestrian exposure or individual-level information about the pedestrians involved, this interpretation should remain cautious.

The main contribution of this study is that it combines smart-city thinking, pedestrian safety literature, open accident data, spatial hotspot analysis, and demographic context. The findings support targeted safety audits, lighting improvements, heavy-vehicle risk management, equity-aware monitoring, and the use of multiple indicators for policy prioritization. A safer smart city should not rely only on digital tools or raw accident counts. It should combine data analysis with local knowledge, inclusive planning, and street-level interventions that improve safety for all pedestrians.

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